We were amazed to realise that Glavkosmos had made and stocked some 252 KVD-1 engines. Here was India, gasping at the idea of developing a cryogenic engine, and there was Russia, the pioneering space super power, in whose warehouses scores of cryogenic rocket engines were gathering dust!

But that was the quintessential Russian style of mass manufacturing. After several interactions with Dunayev and his associates, I learned that the Russians had tested the KVD-1 engines and found them to be of very high specific impulse, but never flew a rocket using one of them. When I asked an engineer, he said with complete nonchalance, 'We never got to.'

I was told the engines were meant for a series of interplanetary missions that Russia was planning. Unimaginable to even the US have been the Russian leaps in space science. Soon after World War II, the Red Army having captured Berlin in 1945, the writing was on the wall: the Soviet Union is going from strength to strength in missile technology and rocketry.

That led to the setting of the cold war a couple of years later. Joseph Stalin took the the Soviet Union to a greater level of industrialization and, even after Nikita Krushchev's de-Stalinisation, Russia remained an American envy when it came to science and technology. How the Soviet Union took the US by surprise with the launch of Sputnik, the first man-made satellite, on 4 October 1957 is now part of the dramatic history of space science.

The US had announced on 2 April 1955 that it would launch a satellite in the International Geophysical Year of 1957-58. Five days later, The Soviet Union said it would also do such a thing 'in the near future'. The first satellite launch by the Soviet Union shocked the US President Dwight D Eisenhower so much that he called it the 'Sputnik moment', a term almost every American President had used to underscore the need for America to wake up to the advances of other countries.

Even as the US was struggling to catch up with the Soviet Union, Russians sent Yuri Gagarin to space on 12 April 1961. America could claim a lead only on 20 July 1969 when Apollo-11 landed Neil Amstrong and Buzz Aldrin on moon. The Soviet Union never got to send a human to the